Isaiah Lindsey

Professor Llovet

ENC2135

8 February, 2026

Investigative Field Essay

"What impacts on American Football will the advancement of machine learning have?"

Introduction

As machine learning models advance, many analysts and organizations in the world of American Football have expressed interest in the possible application of these models. As the ceiling for machine learning models continues to rise, this interest has only grown. The sport of American Football has several facets that these models could assist in reshaping at every level of the sport: injury prevention, talent scouting, decision-making, strategy optimization, etc. Over time, American Football has slowly but consistently shown a willingness to welcome advancements in analytics. Articles dating back decades on analytical decision-making exist and have made a lasting impact on the sport. As an example, an article from 2013 uses analytics to bring up a possible aspect of decision-making at the professional level while down 14 points: "DOWN 14: As with the down-21 case, the obvious answer (kick twice and go to OT) is wrong. If you kick both, your win probability is P(WOT). If you go for two the first time, either you make it and kick the second to win (P(2)) or miss and go for two again to try to force overtime ((1-P(2))*P(2)*P(WOT))." Machine learning is the next step in the realm of sports analytics, and it is in the early stages of integration. The significance of these developments is not insignificant, as the NFL is among the largest entertainment industries in the United States, and its reach is only expanding with international games and the introduction of Olympic Flag Football.

In the current state, machine learning models are unpolished, unclear, and have limited potential. Due to this, trust in them is lacking, which has created hesitance from many NFL organizations to invest in their development. At the college level, many projects and studies are emerging and have made strides in the realistic application of machine learning models. These discoveries, despite being minimal and unpolished, have already cascaded into further research and created impacts at the junior level. These possibilities create the question, "What impacts on American Football will the advancement of machine learning have?" This question is relevant to my personal career path, as I am currently on track to double major in Computer Science and Sports Management. As a result, I have been drawn toward discussing this area of Sports Analytics. The topic is still in its early stages of development, and there are areas that remain unclear and have lacked significant research, such as black-box algorithms and video analysis.

This paper will outline what is currently known about the possibilities of machine learning, how it could be useful in American Football, and its highest potential utilizing research from my sources. The paper will begin by discussing American Football across all levels and the broad impact machine learning can and will have on them. Following this, each level of American Football—junior, college, and NFL—will be discussed individually, examining the specific ways each level could be impacted along with the highest potential possible.

American Football and Machine Learning

As it stands now, the application of machine learning models is very limited. However, one of the most currently researched and developed uses of machine learning is injury prevention and risk prediction. A recent research article outlined one facet of this implementation by attributing which parts of the body are impacted by contact vs. non-contact injuries: “18 studies (47%) limited their analyses to non-contact injuries (injuries without any direct contact or collision with another individual). Two studies considered only contact injuries: one relating to concussion and one to dental injuries. In 47% of the studies (n=18), contact and non-contact injuries were considered jointly.” This is one of many discoveries made by machine learning related to injury prevention. Data like this can be used to assist trainers and coaches in preventing injuries and responding to them more quickly. Understanding what an injury is when it happens allows for quicker and more effective treatment, which is vital to healing.

The sport of football has some of the highest injury rates of any sport due to the intense physical nature of each game. This is especially relevant in high school football, as it is a common subject of speculation regarding the long-term health and safety concerns that come with playing football at a young age. With continued work on these models, it is expected that over the course of around a decade, “ML is likely to greatly enhance the objectivity of decision making in sports science” and specifically, “enabling the identification of movement-specific injury risk factors.” These statements, made in a journal discussing the implications of machine learning models in sport, reflect hopes of achieving many methods of injury prevention. The application of these models at the high school level is a priority and one of the most impactful things these models can do.

One issue slowing further application of machine learning is the lack of transparency the models have regarding their outputs. When it comes to possible applications in decision-making involving training, strategy optimization, draft selections, and other decisions that require extreme planning and careful approaches, it is not desirable for organizations to blindly trust the answer a machine learning model gives without clear understanding of the inner workings of the model along with its reasoning. This problem stems from the lack of transparency from "black-box" algorithms in particular. These algorithms lack ease of interpretation and, in their current state, are untrustworthy among organizations. However, the development of XAI (Explainable Artificial Intelligence) models aims to bridge the gap between machine learning algorithms and understanding.

A publication by Stefan Kranzinger discusses the development of XAI and the related implications for sports analytics. It is explained that research on XAI is currently "fragmented" as it is still in its early stages. However, the author explains that, with more time and research, "The integration of XAI into applied sports settings offers the potential not only to improve transparency but also to support more informed and trustworthy decision-making in athletic performance, coaching, and talent development." Many different models of XAI are discussed and their development is currently improving. One of the most widely adopted models is the SHapley Additive exPlanations (SHAP) model, which is described as providing "insight into how individual input features contribute to a model's predictions." With increased research and development, this can be implemented into the sport of American Football and allow for easier use of and trust in machine learning models. What SHAP can do specifically is assist in cooperative game theory as well as assigning levels of importance to different aspects of game theory. In American Football, a polished machine learning model that can do this and provide reasoning for its explanations would drastically change how machine learning models are viewed at all levels of the sport.

As these models develop and are able to impact these overarching areas of the sport, the different levels of American Football differ, and the ways these models can assist differ as well.

The level of trust needed in a model at the NFL level would be exponentially greater than at the high school or college level, and the complexity of the sport increases as the competition becomes greater. The importance of injury prevention is higher at the high school level, as athletes are still young and are more susceptible to traumatic brain injuries that alter their lives; thus, models that discover better methods of injury prevention would be especially crucial at this level. As for the college level, developing talent and evaluating talent is difficult as there are thousands of recruits all over the nation. Scouting and deciding which players out of thousands will be the starting eleven on the field on game day is practically impossible to perfect. Models can use algorithms to find and list the strengths and weaknesses of players and work to develop talent on the team, which would impact college player development to a high degree.

NFL/College Implementation

It is widely agreed that the NFL Draft is one of the most impactful events in determining how well a team will perform. The players selected build the foundation for a team's future and drafting is one of the most important things for a team to get right in order to become a successful franchise. Despite this, it is extremely common for NFL organizations to get picks wrong and select players who do not perform at the expected level. This can set back a team heavily, especially if several of their selections in the seven rounds of the draft do not work out. It is also one of the leading causes for general managers to be fired, and why teams with strong draft histories often experience greater success. As such, making the correct and most helpful picks for a team's structure is imperative.

To provide an example, the recent Super Bowl champions, the Seattle Seahawks, in three consecutive drafts hit on several crucial picks: in 2023, Jaxon Smith-Njigba and Devon

Witherspoon; in 2024, Byron Murphy and AJ Barner; in 2025, Grey Zabel and Nick Emmanwori. All of these players were starters who contributed to their Super Bowl victory. It is a constantly debated topic which areas to prioritize in a college player and what aspects of scouting talent should matter most. Positional value, athleticism, production (on-field performance), work ethic, and level of competition faced are some of these areas. This is where the application of machine learning models can change how teams make selections.

Machine learning models can analyze all parts of a student's profile and, using historical data from past draft classes, evaluate how likely a player is to succeed at the NFL level, as well as how well they fit a specific team's scheme. Creating a simple formatted list of a player's strengths and weaknesses, providing comparisons to former prospects, and offering other analytical data makes selections easier and gives more information than many traditional forms of evaluation could provide. In a study done by Akshaj Enaganti, a machine learning system was used to predict the draft positions of players. Two simulations of the 2017 NFL Draft were run using different logic: best player available (BPA) and need-based selection. These simulations included several selections that were either exactly accurate to the 2017 draft or one pick away. The accuracy the model simulations had compared to the true draft was high and demonstrates that the models were capable of making informed and accurate decisions for draft day.

This creates impact toward the college world as well, as player draft selection determines the contract a player will receive upon entering the league. For example, referring to the NFL Operations explanation of contract language, players selected in the first round receive many benefits compared to players selected in later rounds. Put simply, the closer a player is selected to the #1 overall pick, the higher their starting salary will be. A machine learning model that influences draft selections impacts the players themselves and their possible success just as much as it impacts the organizations themselves.

Conclusion

The potential of machine learning models and the benefits they can provide to American Football are extremely high, and the further development of these models will be crucial to improving these facets of the sport. The optimization of draft selections, injury prevention, talent evaluation, and player development are some of the most major areas of American Football that will and can be impacted by machine learning. Altogether, this will not only impact the future of the sport itself in the entertainment industry, but the players, coaches, and all other personnel involved in the processes behind the scenes will feel the impacts and benefits from it.